Sentiment Analysis of Financial Terminology in Classic Literature

Parsa Norouzian S5198372

Abstract

This study examines the sentiment associated with financial terminology in two classic novels: *Great Expectations* by Charles Dickens and *The Great Gatsby* by F. Scott Fitzgerald. By analyzing the emotional framing of frequently used financial terms, this research aims to reveal how societal attitudes towards wealth and finance differ between the 19th and 20th centuries. The study uses computational linguistics tools, specifically sentiment analysis models from Hugging Face, to classify the sentiment of financial terms and compare the results across the two literary works.

1 Introduction

Understanding how financial concepts are portrayed in literature can provide valuable insights into historical and cultural attitudes towards wealth, poverty, and social class. This research focuses on analyzing the emotional framing of financial terminology in two iconic literary works: *Great Expectations* (1861) and *The Great Gatsby* (1925). By using computational sentiment analysis, this study aims to compare the portrayal of financial concepts between the Victorian and Jazz Age periods.

Great Expectations was published during the Victorian era, a time of significant industrialization, expanding British imperialism, and stark socioeconomic divisions. The novel explores themes of wealth, poverty, and social mobility, reflecting the anxieties and moral concerns surrounding economic advancement and class disparity in 19th-century Britain. Charles Dickens, known for his social criticism, used the narrative to critique the societal obsession with wealth and the impact it had on personal integrity and relationships.

In contrast, *The Great Gatsby* was written during the Roaring Twenties, a period of economic prosperity and cultural change in the United States following World War I. This era was marked by rapid industrial growth, consumerism, and the pursuit of the American Dream. F. Scott Fitzgerald captures the extravagance and materialism of the Jazz Age while subtly critiquing the hollow pursuit of wealth and status. The novel serves as a reflection of the era's economic optimism, as well as the underlying disillusionment with the American Dream.

By selecting these two novels, this study aims to explore how literature from distinct historical periods reflects differing societal attitudes toward wealth and finance. Comparing these works allows for a deeper understanding of how economic contexts influence literary portrayals of financial concepts and the moral implications surrounding wealth accumulation.

1.1 Research Question and Hypotheses

Research Question: How do the 10 most frequent financial terms differ in sentiment between *Great Expectations* and *The Great Gatsby*, and what does this reveal about societal attitudes toward wealth and finance in the 19th and 20th centuries?

Hypotheses:

- In *Great Expectations*, financial terms will predominantly exhibit negative or moralistic sentiment, reflecting concerns about poverty, debt, and class struggles common in 19th-century society.

- In *The Great Gatsby*, financial terms will show more positive or neutral sentiment, reflecting materialism and the pursuit of wealth during the economic prosperity of the 1920s.

1.2 Related work

Jockers and Mimno (2013) applied computational methods to examine themes in 19th-century novels, identifying patterns related to social and economic themes. Similarly, Buechel et al. (2018) conducted sentiment analysis on historical texts, demonstrating how emotional language reflects social and economic conditions over time. These studies support the use of computational analysis in understanding how literature reflects societal values.

2 Methodology

2.1 Data Collection

The primary texts analyzed in this study are:

- *Great Expectations* by Charles Dickens (sourced from Project Gutenberg)
- *The Great Gatsby* by F. Scott Fitzgerald (sourced from Project Gutenberg)

2.2 Data Processing

1. Extracted the 10 most frequent financial terms from each novel.
2. Retrieved all sentences containing these financial terms.
3. Applied Hugging Face's sentiment analysis model to classify each sentence as Positive, Negative, or Neutral.
4. Calculated sentiment distribution percentages for each term.
5. Visualized the results using bar charts.

3 Results

3.1 Sentiment Analysis of *Great Expectations*

The sentiment analysis of financial terms in *Great Expectations* revealed the following distribution:

- Positive Sentiment: **42.60%**
- Negative Sentiment: **57.40%**

3.2 Sentiment Analysis of *The Great Gatsby*

The sentiment analysis of financial terms in *The Great Gatsby* revealed the following distribution:

- Positive Sentiment: **46.48%**

Term	Positive	Negative
Money	20	56
Poor	24	50
Account	14	37
Fortune	21	19
Return	85	86
Bank	15	23
Rich	30	15
Interest	19	21
Capital	5	9
Loss	6	6

Table 1: Sentiment distribution of financial terms in *Great Expectations*

Term	Positive	Negative
Money	8	13
Rich	4	4
Poor	5	4
Interest	8	7
Bond	2	6
Share	2	1
Wealthy	1	2
Penniless	1	1
Privileged	1	0
Finance	1	0

Table 2: Sentiment distribution of financial terms in *The Great Gatsby*

- Negative Sentiment: **53.52%**

These results indicate that *Great Expectations* predominantly frames financial concepts in a negative light, reflecting 19th-century societal concerns about poverty and class struggles. In contrast, *The Great Gatsby* shows a more balanced sentiment distribution, consistent with themes of materialism and wealth during the 1920s.

Discussion The results highlight distinct differences in how financial terminology is emotionally framed across the two literary works. *Great Expectations* exhibits a stronger negative sentiment toward financial terms, which aligns with the novel's focus on poverty, social mobility, and the moral consequences of wealth. This suggests that 19th-century literature often portrayed finance through a lens of social critique and moral responsibility.

In contrast, *The Great Gatsby* demonstrates a more balanced sentiment, with slightly more positive depictions of wealth-related terms. This reflects the cultural and economic optimism of the 1920s, where wealth was associated with success, ambition, and social status. However, the presence of negative sentiment also indicates underlying critiques of materialism and the eventual consequences of financial excess.

These results support the hypothesis that societal and historical contexts shape the emotional framing of financial concepts in literature. This analysis highlights how literature serves as a reflection of contemporary social and economic values.

4 Conclusion

The sentiment analysis confirms that financial terminology in literature reflects broader societal values. *Great Expectations* portrays financial topics more negatively, aligning with themes of economic hardship and social mobility in the Victorian era. *The Great Gatsby*, on the other hand, demonstrates a more neutral or slightly positive portrayal, reflecting the materialistic culture of the Jazz Age. However, the difference in sentiment between the two novels is not significant, suggesting that further and more detailed research is necessary to better understand the relationship between literary framing of finance and historical context.

References

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